Predictive Learning — Homework 2

VC-theory and its Applications

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Problem 1 (model selection) and Problem 2 (trading international mutual funds)

Code and data: https://github.com/tc9305/predict_learning/tree/main/HW2

Problem 1: Model selection for trigonometric polynomial regression

Experimental setup

I follow the procedure of Section 4.4 and compare analytic model selection criteria for univariate regression with trigonometric polynomials

$$f_m(x) = w_0 + \sum_{k=1}^{m} w_k \cos(2\pi k x), \qquad \text{DoF} = m + 1,$$

fitted by ordinary least squares. I draw inputs from $x \sim U[0, 1]$ and use two data sets:

- **Pure noise:** $y \sim N(0, 1)$. The true function is $f(x) = 0$, so the correct model is a constant (DoF = 1).
- **Sine-squared:** $y = \sin^2(2\pi x) + \varepsilon$, $\varepsilon \sim N(0, 0.2^2)$. Since $\sin^2(2\pi x) = \frac{1}{2} - \frac{1}{2}\cos(4\pi x)$, the target belongs to the model with $m = 2$, and the correct DoF is 3.

I run 300 independent realizations for $n = 30$ (candidate DoF 1–20) and $n = 100$ (candidate DoF 1–25). In each realization, every method selects a DoF. I refit the model with that DoF and measure its *prediction risk* as the mean squared error between the fitted model and the *noise-free* target on 2000 independent test points. No model can predict the noise, so this risk measures the distance between the selected model and the truth.

Model selection methods

Training error falls as complexity grows, so it cannot guide model selection by itself. In one sine-squared sample with $n = 30$, the training MSE fell from 0.107 at DoF = 1 to 0.0067 at DoF = 20, below the noise variance $\sigma^2 = 0.04$. Each analytic method multiplies the training error R_{emp} by a penalization factor $r(p)$, where $p = \text{DoF}/n$, and selects the DoF that minimizes $r(p)\, R_{\text{emp}}$:

Method	Penalization factor $r(p)$	Reference
Final prediction error (fpe)	$(1 + p)/(1 - p)$	Eq. (2.11)
Schwartz criterion (sc)	$1 + (\ln n)\, p/(1 - p)$	Eq. (2.12)
Generalized cross-validation (gcv)	$(1 - p)^{-2}$	Eq. (2.13)
VC bound (vc)	$\left(1 - \sqrt{p - p\ln p + \ln n/(2n)}\right)_+^{-1}$	Eq. (4.15)

For vc, I set the factor to infinity when the term under the square root reaches 1, which excludes that model. At $p = 0.1$ the four factors lie between 1.2 and 1.5. At $p = 0.5$ with $n = 30$ they spread out: fpe gives 3.0, gcv 4.0, sc 4.4, and vc 20.2.

The fifth method, leave-one-out cross-validation (cv), estimates prediction risk from held-out data. I compute the leave-one-out error from a single fit with the leverage formula $e_i/(1 - h_{ii})$, where e_i is the ordinary residual and h_{ii} the i-th diagonal entry of the hat matrix. In my checks it matched the result of n explicit refits to four decimal places.

Results

Figures 1 and 2 present the results in the style of Figs. 4.10–4.11. Each box spans the 25th to 75th percentiles with a line at the median, and the whiskers reach the 5th and 95th percentiles. The risk panels use a logarithmic scale, and a dashed line in the DoF panels marks the correct DoF.

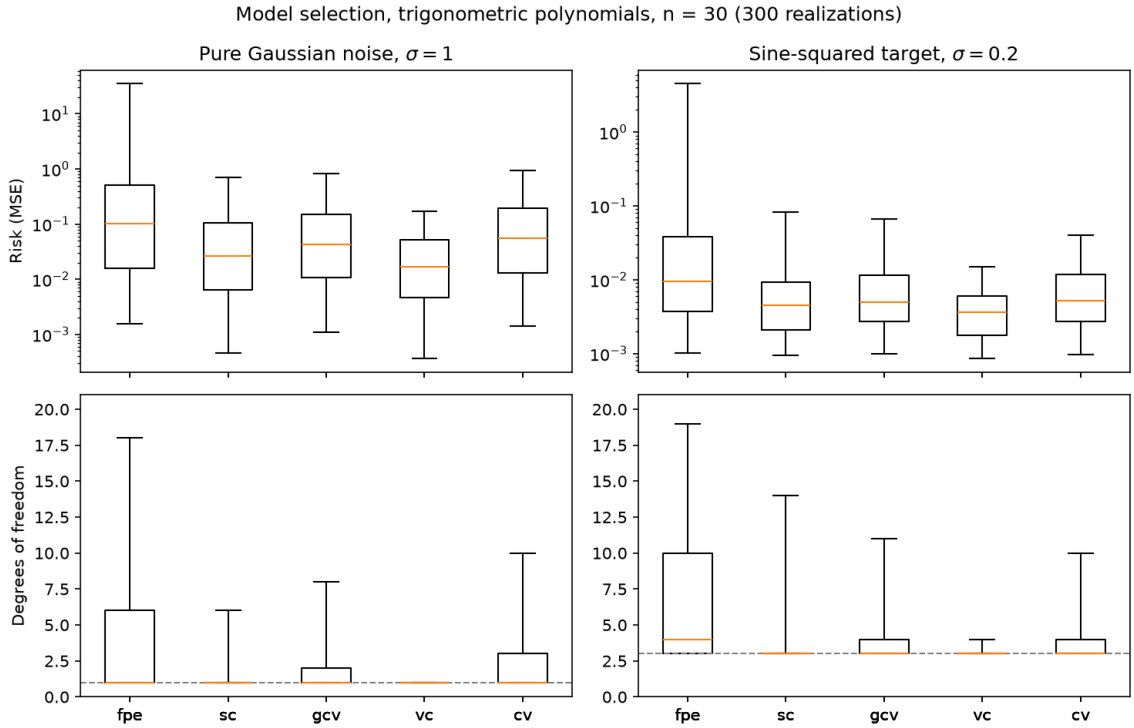


Figure 1: Model selection results for $n = 30$ (300 realizations). Top: prediction risk (log scale); bottom: selected DoF. Left: pure noise; right: sine-squared.

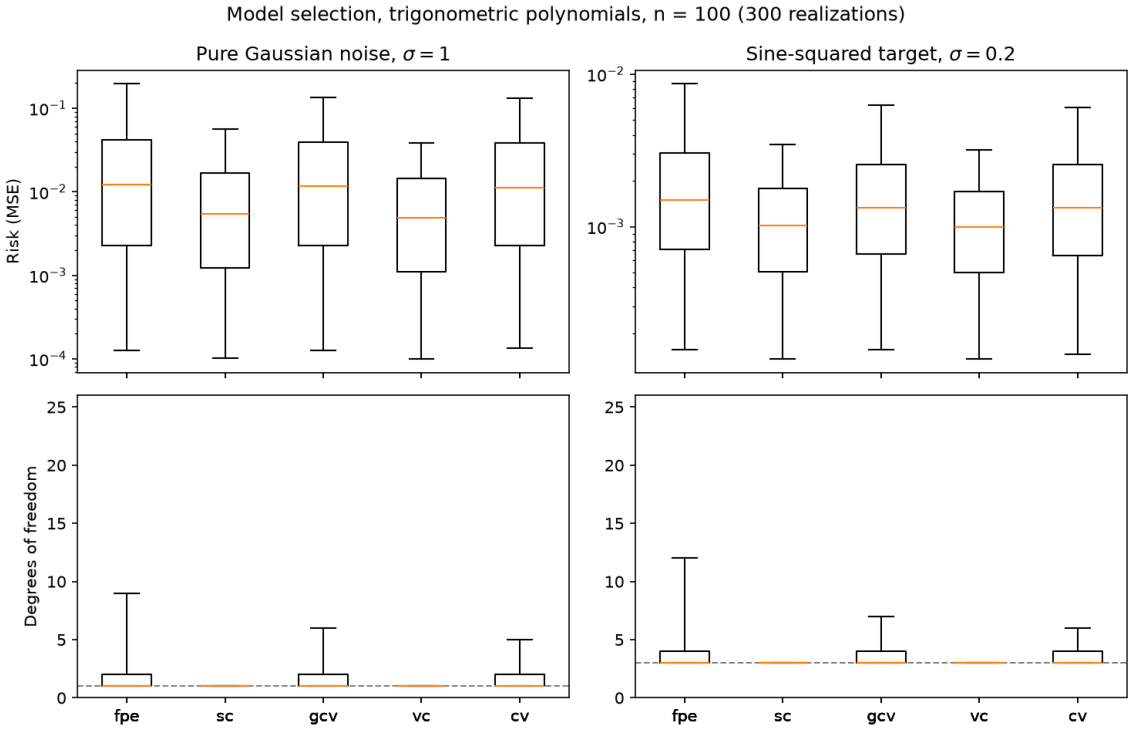


Figure 2: Model selection results for $n = 100$ (300 realizations), same layout as Figure 1.

	$n = 30$				$n = 100$			
	Pure noise		Sine-squared		Pure noise		Sine-squared	
Method	correct	risk p95	correct	risk p95	correct	risk p95	correct	risk p95
fpe	53.0%	35.7	46.3%	4.54	73.0%	0.197	68.7%	0.0087
sc	82.7%	0.708	76.0%	0.0821	95.3%	0.0568	97.0%	0.0035
gcv	69.0%	0.831	67.7%	0.0672	74.0%	0.134	73.0%	0.0063
vc	97.7%	0.170	92.3%	0.0149	100.0%	0.0381	99.3%	0.0032
cv	64.7%	0.945	64.7%	0.0402	74.7%	0.132	74.3%	0.0060

correct: share of the 300 realizations that selected the true DoF; *risk p95*: 95th percentile of the prediction risk.

Small samples ($n = 30$). The share of correct selections follows the strength of the penalty: vc (97.7% for pure noise, 92.3% for sine-squared) > sc > gcv > cv > fpe (53.0%, 46.3%). For sine-squared, fpe is the one method with a wrong median DoF (4 instead of 3). Its light penalty lets a large drop in training error win: in one sine-squared sample, fpe chose DoF = 18 while the other four methods chose DoF = 3. The tails of the risk distribution show the largest gaps. For pure noise, the 95th-percentile risk of fpe (35.7) is about 200 times that of vc (0.17), while their medians differ by a factor of six (0.103 vs. 0.017). In its worst 5% of realizations, fpe picked models with DoF 11–20, which swing far from zero between the training points; the worst vc selections had DoF 1 or 2.

Effect of sample size ($n = 100$). With $n = 100$, every method improves. vc selects the correct DoF in 100% (pure noise) and 99.3% (sine-squared) of realizations, and sc in 95.3% and 97.0%. fpe, gcv, and cv level off at 70–75%. For small p, these three criteria charge about $2\sigma^2/n$ for each extra parameter, and a spurious term beats that charge with a probability that does not shrink as n grows. The sc charge, about $(\ln n)\,\sigma^2/n$, grows with n. The 95th-percentile risk of fpe falls from 35.7 to 0.197 as its 95th-percentile DoF drops to 9 (pure noise) and 12 (sine-squared).

Conclusion

Across both data sets and both sample sizes, the VC penalization chose the correct model most often and had the lowest 95th-percentile risk. Its lead is widest at $n = 30$, where a wrong choice costs the most. The five methods have similar median risks and separate in the worst 5% of realizations, so a fair comparison needs the full risk distribution.

Problem 2: Trading international mutual funds

Data and encoding

I use daily closing prices of the S&P 500 (`^GSPC`) and the Fidelity International Discovery fund (FIGRX) from Yahoo Finance, and the euro exchange rate (euros per US dollar) from the UBC Pacific Exchange Rate Service. Following the course preprocessing note, I keep the dates present in all three series: 243 trading days in 2004 and 245 in 2005. Alignment removes nine 2004 days on which the UBC file has no quote, most of them Canadian holidays, and seven US market holidays on which it does.

The inputs are today's percentage changes, x_1 for the S&P 500 and x_2 for the exchange rate, each measured against the previous aligned day. The output y is tomorrow's percentage change of FIGRX. I join the price series across years to compute the changes at the year boundaries. The output for December 31, 2004 uses the first aligned day of 2005 (January 4). The input for January 6, 2004, the first aligned day of that year, needs the close of December 31, 2003, which the provided files do not contain; I retrieved the three values from the same sources (S&P 500 1111.92, FIGRX 23.92, 0.79386 euros per dollar). I obtain 243 training samples (January 6 to December 31, 2004) and 244 test samples (January 4 to December 29, 2005). The last day of 2005 has no next-day output.

FIGRX closed unchanged on seven 2004 days. I drop those days when I fit the quadratic classifier and when I compute accuracy. The trading simulation keeps them: a zero change produces zero gain under either position.

Classifiers

(a) Linear classifier. I fit $t(\mathbf{x}) = w_0 + w_1 x_1 + w_2 x_2$ to the real-valued outputs by least squares and predict UP when $t(\mathbf{x}) > 0$. The 2004 data give

$$t(\mathbf{x}) = 0.0469 + 0.0844\,x_1 - 0.0073\,x_2.$$

The positive S&P 500 coefficient fits the view that foreign markets follow the US market with a one-day lag. The exchange-rate coefficient is close to zero. With a positive intercept, the rule predicts DOWN after the S&P 500 falls by more than about 0.56%, and it predicts UP on 79% of training days.

(b) Quadratic classifier. I convert the outputs to class labels $y = \text{sign}(y)$ and fit

$$g(\mathbf{x}, \mathbf{w}) = w_1 x_1 + w_2 x_2 + w_3 x_1^2 + w_4 x_2^2 + w_5 x_1 x_2 + w_0$$

to the ± 1 labels by least squares, then predict UP when $g > 0$. The assignment leaves the estimation method open; least squares keeps the two approaches comparable. The fitted coefficients are $w_0 = 0.1594$, $w_1 = 0.0787$, $w_2 = 0.1112$, $w_3 = -0.0606$, $w_4 = -0.1470$, and $w_5 = 0.1361$. Both squared terms are negative and the intercept is positive, so the boundary is an ellipse with the UP region inside. The model predicts UP on calm days and DOWN after large moves in either input.

Classifier	Training accuracy (2004)	Test accuracy (2005)
Linear	51.7%	57.1%
Quadratic	57.2%	55.8%
Always UP	52.5%	55.0%

The quadratic model gains 5.5 points of training accuracy over the linear one and loses 1.3 points on the test year, the overfitting pattern I saw in Problem 1. With about 240 days per year, one day moves accuracy by 0.4 points, and the standard error of an accuracy estimate is close to 3 points. Neither classifier beats the always-UP rule by more than that margin.

Decision boundaries

Figure 3 plots both boundaries over the training and test data in the required $(-2\%, +2\%)$ range. The linear boundary runs close to vertical at $x_1 \approx -0.56\%$, because the exchange rate carries almost no weight. The quadratic ellipse tilts toward the upper right and extends past the plotted range. Its lower edge near $x_2 = -0.7\%$ wraps around a group of 2004 DOWN days that has no counterpart in 2005. In all four panels the UP and DOWN days overlap, and no boundary of either shape separates them.

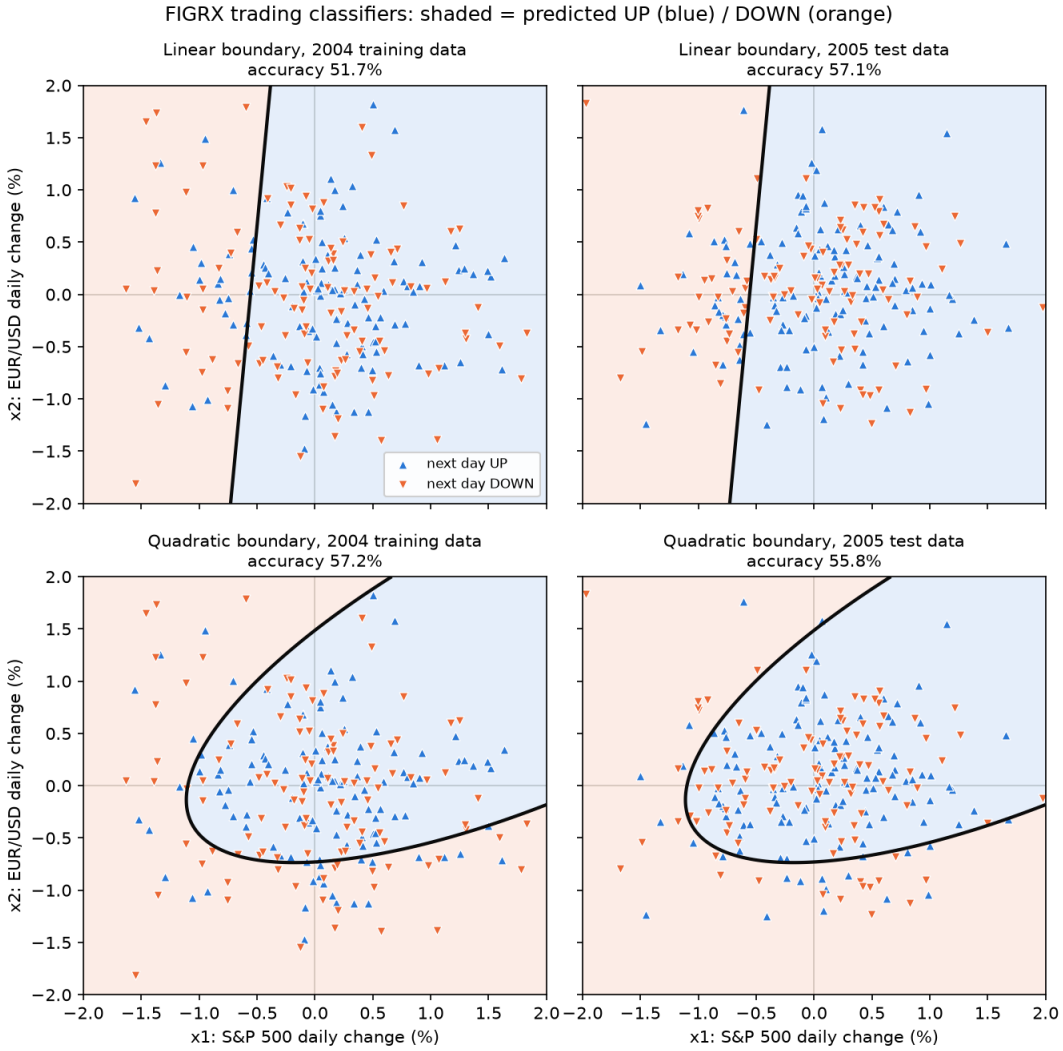


Figure 3: Decision boundaries of the linear (top) and quadratic (bottom) classifiers over the 2004 training data (left) and the 2005 test data (right). Shading shows the predicted class; markers show the actual next-day FIGRX change.

The data give the ellipse little support. On 2005 days inside it, FIGRX rose the next day 56.6% of the time, against 47.7% outside. The outside group holds 44 days, so its standard error is about 7.5 points, and the gap falls within sampling error. In 2004 the same comparison gave 56.7% and 41.5%, a contrast the fit itself produced.

Trading results

Each strategy holds FIGRX on days its classifier predicts UP and cash otherwise. Starting from $1 million, I compound the daily returns over the 2005 test period. GAIN is the cumulative return at the end of the year, and EXPOSURE is the share of days invested in FIGRX.

Strategy	GAIN	EXPOSURE
Linear	21.81%	76.6%
Quadratic	15.19%	82.0%
Buy-and-Hold	14.92%	100.0%

Buy-and-Hold earns 14.92%, from the January 4 close (27.55) to the December 30 close (31.66). The linear strategy earns 6.9 points more while staying in cash on 57 days. On those days FIGRX moved −0.10% on average, against +0.11% on the days the strategy held the fund. Accuracy counts each day the same; the gain depends on which days the strategy avoids. To judge luck, I drew 10,000 random strategies invested on the same 76.6% of days. Their median gain is 11.2%, and 3.1% of them reach 21.81% or more.

Discussion

(a) Can this strategy be used in practice? I would not trade my own money with it. The 2005 result rests on one test year, and I picked the better of two models after seeing both. Neither classifier beats the always-UP rule in accuracy by more than the sampling error, and the UP and DOWN days overlap across the input plane. The linear rule profits from the stale pricing of international funds, whose prices use foreign closes that have not yet reacted to the US market. After the 2003 mutual-fund scandal, fund companies adopted fair-value pricing, and many restrict short-term trading or charge redemption fees, which would close this gap. The assignment also assumes zero transaction costs. The linear strategy switches position 89 times in 2005; at 0.1% per switch, costs would take about 8.5% and erase its 6.9-point lead.

(b) VC-theoretical explanation. Both models are small for the sample size. The linear classifier has $h = 3$ parameters and the quadratic one $h = 6$, so $p = h/n$ equals 0.012 and 0.025 with $n = 243$. At these values the VC factor of Eq. (4.15) is 1.39 and 1.56, compared with 20.2 at $p = 0.5$ and $n = 30$ in Problem 1, and the training error gives a usable estimate of the test error. Complexity control has little to correct here, which is why these methods skip it. The training accuracy itself sits near the level of random guessing (51.7% and 57.2%), and a VC bound built on that error promises no useful performance whatever complexity I choose. The weak link between today's inputs and tomorrow's FIGRX change limits both classifiers, and a larger or smaller model cannot repair it.